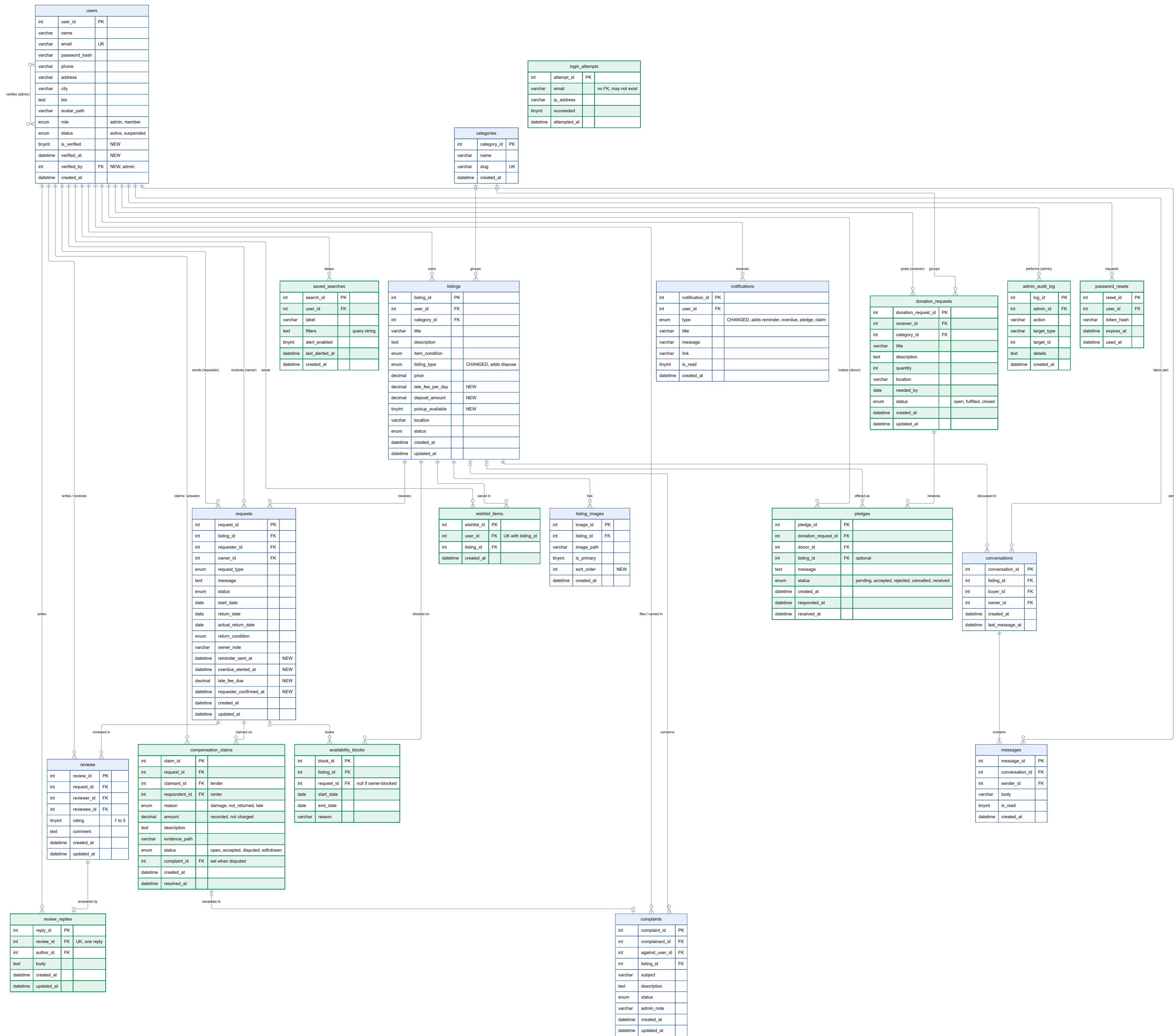


NexUse — Entity–Relationship Diagram

Full system: the 10 tables built for the interim, plus the 10 tables and new columns the planned features need · CS-31 · September 2026



 Built (interim)
 Planned table
 "NEW" planned column
 "CHANGED" extended enum

Tables

Table	Status	Feature (interim report §6.3)
users	Built · extended	+ verified badge (task 8)
categories	Built	—
listings	Built · extended	+ late fee (4), deposit and pickup flag (14), dispose type (11)
listing_images	Built · extended	+ photo order
requests	Built · extended	+ reminders and overdue alerts (2), late fee due (4), renter / receiver confirmation (10)
reviews	Built	—
complaints	Built	Target of disputed claims (3)
notifications	Built · extended	+ reminder, overdue, pledge, claim types
conversations	Built	—
messages	Built	—
donation_requests	Planned	Donation requests (task 1)
pledges	Planned	Pledges (task 1)
compensation_claims	Planned	Compensation claims (task 3)
availability_blocks	Planned	Availability calendar (task 12)
review_replies	Planned	Reply to reviews (task 9)
wishlist_items	Planned	Wishlist (task 11)
saved_searches	Planned	Saved searches with alerts
password_resets	Planned	Self-service password reset (task 13)
login_attempts	Planned	Sign-in rate limiting (task 15)
admin_audit_log	Planned	Audit log of admin actions (task 15)

Design rules

- Every table has a surrogate integer primary key; relationships are enforced by foreign keys.
- A **request** links a listing to both its requester and its owner. Reviews, compensation claims and availability bookings all hang off a request.
- A **pledge** answers a donation request and may point at an existing donate listing.
- A disputed **compensation claim** becomes a complaint, so admins handle one queue.
- Amounts (price, late fee, deposit, claim) are **recorded, never charged** — payments stay outside the platform.
- Derived figures — average rating, unread counts, admin reports — are calculated by query, not stored.
- **login_attempts** keeps the typed email rather than a foreign key, because failed attempts may be for an address with no account.

Crow's-foot notation. PK primary key, FK foreign key, UK unique. Admin reports are calculated from these tables by query and need no table of their own. Payments and delivery stay out of scope, so no table records them.